

HISTOPATH

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(I)
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(II)

DECLARATION

We hereby declare the following usage of the open source code and prebuilt libraries in our minor project in **Semester V** with the consent of our supervisor. We also measure the similarity percentage of pre written source code and our source code and the same is mentioned below. This measurement is true with best of our knowledge and abilities.

Place:

Date: 22nd November 2025

Enrollment Number	Student Name	Student Signature
2310382	Prish Keshari	
23103077	Akshit Gupta	
2303022	Arjun Gupta	

1. LIST OF LIBRARIES

DEEP LEARNING

1. TensorFlow / Keras (tensorflow, tensorflow.keras)

Used for:

- Building the Convolutional Neural Network (CNN) model in the notebook
- Training the model (layers, optimizers, callbacks)
- Loading the saved model (load_model) in FastAPI for inference
- Performing predictions on uploaded images

2. h5py

Used internally to:

- Save and load model weights stored in .h5 format
 - Required because model.h5 / custom.h5 uses the HDF5 file structure
-

WEB BACKEND

1. FastAPI

Used for:

- Creating the API server
- Handling image uploads (UploadFile)
- Defining endpoints (e.g., /predict)
- Sending prediction results back to the frontend

2. Uvicorn

Used as:

- The ASGI server that runs the FastAPI app

- Starts the backend on localhost:8000
-

IMAGE PROCESSING

1. Pillow (PIL)

Used for:

- Opening raw uploaded image bytes
 - Converting images to RGB
 - Resizing images to model input size (96×96)
 - Converting image → NumPy array
-

DATA SCIENCE / ML

1. NumPy

Used for:

- Converting processed images into arrays
- Expanding batch dimensions
- Normalizing pixel values (dividing by 255)
- General numerical handling in the notebook

2. Pandas

Used for:

- Handling dataset metadata
- Loading CSVs (labels, training info)
- Any DataFrame operations used during analysis

3. Scikit-learn

Used for:

- Model evaluation:
 - Classification report
 - Confusion matrix
 - ROC curve & AUC score
- Splitting data or performing preprocessing (if used)

4. Matplotlib

Used for:

- Plotting training accuracy/loss curves
- Visualizing sample images
- Visualizing confusion matrix

5. Seaborn

Used for:

- Creating heatmap-style plots
 - Enhancing visualizations (confusion matrix heatmap)
-

BACKEND (NODE)

1. NextServer

Used to:

- Provides NextResponse for sending API responses
- Enables building server API routes in Next.js

2. retrieveContext (custom)

Used to:

- Performs RAG (context retrieval)
 - Fetches relevant chunks from stored documents
-

FRONTEND (REACT + NEXT.JS)

1. React

- Builds UI components (Chatbot, ImageAnalyzer, etc.)
- Manages component state with hooks
- Handles events & rendering

2. Next JS

- Renders pages and layouts
 - Provides metadata management
 - Handles routing and server components
 - Loads the Inter font
 - Provides optimized Google Font handling
-

2. PRE-BUILT FEATURES IN LIBRARIES OR SOURCE CODE

- Tailwind CSS
- Conv2D
- MaxPooling2D
- Dense
- ImageDataGenerator
- model.predict
- FastAPI

3. PERCENTAGE OF PRE-WRITTEN SOURCE CODE VS CUSTOM CODE

Source Category	Approx. Code Contribution
Pre-built/Open-source Libraries	35%
Custom Features Developed by You	65%

(III)
CERTIFICATE

This is to certify that the work titled “**HistoPath**” submitted by **Arjun Gupta, Prish Keshari, Akshit Gupta** in partial fulfilment for the award of degree of B.Tech in Computer Science of Jaypee Institute of Information Technology, Noida has been carried out under my supervision. This work has not been submitted partially or wholly to another University or Institute for the award of this or other degree or diploma

Signature of Supervisor:

Name of Supervisor: Dr. Amit Mishra

Date: 22nd November, 2025

(IV)

ACKNOWLEDGEMENT

While bringing out this report to its final form, we came across a number of people whose contributions in various ways helped our field of research and they deserve special thanks. It is a pleasure to convey our gratitude to all of them. First and foremost, we would like to express our deep sense of gratitude and indebtedness to our supervisor **Amit Mishra**, their invaluable encouragement, suggestions and support from an early stage of the research and providing me extraordinary experiences throughout the work. Above all, their priceless and meticulous supervision at each and every phase of work inspired us in innumerable ways. We specially acknowledge them for their advice, supervision, and the vital contribution as and when required during this research. Their involvement with originality has triggered and nourished our intellectual maturity that will help us for a long time to come. We are proud to record that we had the opportunity to work with an exceptionally experienced Professor like him.

(V)

SUMMARY

HistoPath is an AI-powered web application designed to assist in the analysis of histopathological images for metastatic cancer detection. The system integrates a FastAPI backend for image processing and deep-learning inference with a modern Next.js frontend that provides an interactive user interface.

Users can upload histopathology slide images, which are processed through a custom-trained Convolutional Neural Network (CNN) model. The backend preprocesses the image, performs prediction, and returns both the classification result (Metastatic / Non-Metastatic) and the associated confidence score. This output is then combined with a contextual chat assistant called Histopath, which uses Retrieval-Augmented Generation (RAG) and Groq's LLMs to answer user queries based on image findings and domain-specific documents.

The platform ensures that the assistant remains strictly informational—avoiding medical diagnosis—while helping users understand histopathology concepts, metastasis indicators, and the AI model's output.

Overall, HistoPath provides a seamless integration of machine learning, web development, and AI-assisted conversation to create a powerful educational and research-support tool for cancer metastasis detection.

CHAPTER 1

INTRODUCTION

1.1 General Introduction

Histopathology is a critical branch of medicine focused on the microscopic examination of biological tissues to study the manifestations of disease. It serves as the gold standard for diagnosing various conditions, particularly cancer. The traditional workflow involves obtaining a biopsy, processing the tissue into thin slices on glass slides, staining them (typically with Hematoxylin and Eosin), and examining them under a microscope to identify abnormalities.

A key challenge in cancer prognosis is the detection of **metastasis**, which is the spread of cancer cells from the primary tumor to other parts of the body, such as the lymph nodes. Metastasis significantly alters the staging and treatment plan for a patient. However, manual detection of metastatic regions in whole-slide images (WSIs) is laborious, time-consuming, and subject to inter-observer variability among pathologists. The **HistoPath AI** project aims to automate this process using Deep Learning to assist pathologists in making faster and more accurate diagnoses.

1.2 Problem Statement

The primary challenge in cancer prognosis is the accurate and timely detection of **metastasis**, the spread of cancer cells from the primary tumor to surrounding tissues, particularly the lymph nodes. This diagnosis is crucial for staging cancer and determining the appropriate treatment plan.

Currently, the "gold standard" for this diagnosis involves **histopathology**, where pathologists manually examine biological tissue samples under a microscope. This process presents several significant difficulties:

- **Labor-Intensive and Time-Consuming:** Pathologists must meticulously scan gigapixel whole-slide images (WSIs) to identify minute regions of tumor tissue, a task that is mentally and visually exhausting.
- **Subjectivity and Variability:** Manual diagnosis is subject to inter-observer variability, where different pathologists may interpret the same slide differently, leading to potential inconsistencies in diagnosis.
- **High Volume of Data:** The increasing volume of histopathological images exacerbates the workload, potentially delaying critical diagnoses.
- **Need for Precision:** Missing even a small area of metastatic tissue (micro-metastasis) can have severe consequences for patient outcomes.

Therefore, the core problem is to develop an automated, reliable, and efficient system that can assist pathologists by accurately classifying histopathologic image patches as either "Metastatic" or "Non-Metastatic," thereby streamlining the diagnostic workflow.

1.3 Significance of the Problem

The significance of automating metastasis detection extends beyond simple efficiency; it has profound implications for patient care and the medical field.

- **Enhanced Diagnostic Accuracy:** By leveraging Deep Learning, specifically Convolutional Neural Networks (CNNs), the system can identify subtle patterns in tissue architecture that might be missed by the human eye, reducing false negatives.
- **Reduced Pathologist Workload:** Automation acts as a "second pair of eyes," filtering out obvious non-cancerous regions and highlighting suspicious areas. This allows pathologists to focus their expertise on complex cases, significantly reducing their cognitive load and fatigue.
- **Faster Turnaround Times:** Automated pre-screening can accelerate the diagnostic process, enabling faster initiation of treatment protocols which is vital for cancer patients.
- **Standardization:** An AI-based system provides consistent and reproducible results, helping to standardize diagnostic criteria across different pathologists and institutions.
- **Clinical Accessibility:** Unlike many "black box" research models, this project introduces a novel, end-to-end web platform. It features a **Chatbot Assistant** to guide non-technical medical staff and an **Image Analyzer** with interpretable heatmaps (Grad-CAM). This bridge between advanced AI and clinical utility fosters trust and facilitates the adoption of these tools in real-world medical settings.

1.4 Empirical Study

To validate the practical necessity of our automated solution, it is critical to understand the current constraints of the clinical environment. The demand for accurate histopathological diagnosis is outpacing the available workforce, creating a "diagnostic gap" that AI is uniquely positioned to bridge.

- **Workforce Crisis:** The country has seen an approximately **19.5% decrease** in the active pathologist workforce between 2010 and 2025, despite a concurrent rise in cancer cases. This has led to a significant increase in the caseload per pathologist.
- **Diagnostic Discordance:** Manual diagnosis is subject to human variability. Studies have shown that inter-observer variability in histopathology can be significant, with diagnostic concordance rates for certain complex cases dropping below **75-80%**. For lymph node metastasis specifically, the detection of micro-metastases (<2mm) is notoriously difficult and prone to being missed during standard manual review.
- **Time Efficiency:** A pathologist typically spends minutes reviewing a single Whole Slide Image (WSI). With digital pathology adoption growing—projected to reach a market size of **\$4.2 billion by 2035**—there is a critical need for tools that can pre-screen these images. AI assistance has been shown to potentially reduce review time by highlighting regions of interest, allowing pathologists to focus on high-value decision-making.

Against this backdrop of diminishing resources and increasing demand, our empirical evaluation focuses on the model's ability to deliver high-precision results that can alleviate these pressures.

1.5 Brief Description of the Solution Approach

The solution employs a Deep Learning approach centered on **Convolutional Neural Networks (CNNs)**, which are highly effective for image recognition tasks.

- **Model Architecture:** A custom CNN architecture was designed to extract hierarchical features from the 96x96 pixel pathology patches. The network likely includes convolutional layers for feature extraction followed by dense layers for binary classification.
- **Training Pipeline:** The model utilized the TensorFlow/Keras framework. Preprocessing involved normalizing image pixel values and potentially data augmentation to improve generalization.
- **Deployment:** The trained model (model.h5) is deployed via a web interface built with **Next.js** and **React**. The backend handles image uploads, runs the model inference, and returns predictions to the user.
- **User Interface:** Features include an intuitive header, theme switcher for accessibility, and a dedicated chatbot component to assist users in navigating the tool.

1.6 Comparison of the Existing Approaches to the Problem Found

1. Speed & Efficiency	Slow. Pathologists must manually scan gigapixel Whole Slide Images (WSIs) at high magnification, a process that can take 15-20 minutes per difficult case.	Rapid. The CNN model processes image patches in milliseconds, enabling near-instantaneous pre-screening of complex slides.	Drastically reduces the "time-to-diagnosis," allowing patients to start treatment sooner.
2. Consistency	Variable. Diagnosis is subjective. "Inter-observer variability" means two pathologists might disagree on the same slide (concordance can be as low as 75% for difficult cases).	Standardized. The model provides deterministic, reproducible predictions based on learned features, eliminating subjective bias.	Ensures every patient receives a diagnosis based on the same rigorous standard, regardless of location or time.
3. Fatigue & Vigilance	Limited. Human performance degrades with fatigue. After hours of reviewing slides, the risk of missing small "micro-metastases" increases.	Unlimited. The AI maintains 100% attention span 24/7, scrutinizing every pixel with the same level of accuracy.	Eliminates error-prone fatigue, acting as a "safety net" that never gets tired.
4. Scalability	Constrained. Dependent on a shrinking workforce (17.5% drop in pathologists vs. rising cancer rates). You cannot simply "add more humans" quickly.	Infinite. The software can be deployed instantly to thousands of clinics via the web platform, scaling effortlessly to meet demand.	Solves the critical workforce bottleneck, making expert-level screening available even in resource-poor regions.

CHAPTER 2

LITERATURE SURVEY

2.1 Summary of Papers Studied

1. [2015] An Introduction to Convolutional Neural Networks

- **Authors:** Keiron O'Shea and Ryan Nash
- **Relevance to HistoPath:** This paper establishes the fundamental theory behind the Convolutional Neural Networks (CNNs) used in your project. It explains why CNNs are superior to traditional Artificial Neural Networks (ANNs) for image analysis, specifically for your 96x96 pixel patches.
 - **Spatial Architecture:** It details how CNNs preserve the spatial structure of images by arranging neurons in three dimensions (width, height, depth), which is crucial for analyzing your RGB histopathology slides.
 - **Feature Extraction:** The paper explains the role of convolutional layers as feature extractors. In your project, these layers are responsible for learning filters that detect edges, textures, and cellular shapes indicative of metastatic tissue.
 - **Efficiency:** It highlights parameter sharing, a mechanism that allows your model to learn efficient representations without the computational explosion that would occur if every pixel in your patch were fully connected to every neuron. This efficiency enables your model to be deployed effectively on a web platform.

2. [2024] Histopathological Cancer Detection Based on Deep Learning and Stain Images

- **Authors:** Dina M. Ibrahim, Mohammad Ali A. Hammoudeh, and Tahani M. Allam
- **Relevance to HistoPath:** This study addresses a common real-world challenge your project faces: variability in tissue appearance. It provides theoretical support for preprocessing steps that could enhance your model's robustness.
 - **Stain Normalization (SN):** The authors explore the theory that variations in Hematoxylin and Eosin (H&E) staining (due to different lab protocols or scanner types) can confuse a model. They propose Stain Normalization as a critical preprocessing step to standardize colors before training, ensuring the model learns to detect cancer based on tissue structure rather than color intensity.

- **Architecture Comparison:** The paper provides a theoretical basis for choosing deeper architectures like ResNet or DenseNet over shallower ones. It explains how mechanisms like "skip connections" (ResNet) and "dense connections" (DenseNet) allow models to learn more complex, non-linear features from medical images without suffering from vanishing gradients.

3. [2023] Histopathologic Cancer Detection

- **Authors:** Varan Singh Rohila, Neeraj Lalwani, and Lochan Basyal
- **Relevance to HistoPath:** This paper is highly relevant as it specifically utilizes the PatchCamelyon benchmark dataset (the same dataset used in your project) ¹ to establish theoretical baselines for cancer prognosis.
 - **MLP vs. CNN Theory:** The authors theoretically demonstrate why a Convolutional Model is inherently better suited for histopathology than a Multi-Layer Perceptron (MLP). They explain that MLPs require flattening the image, which destroys spatial information, whereas CNNs use filters to create feature maps that preserve the location and strength of detected features (like tumor cells)².
 - **Ensemble Approaches:** The study explores two theoretical approaches to Ensemble Learning:
 - *Hard Ensemble (Majority Vote):* Combining the discrete output decisions of multiple models (e.g., ResNet50 and InceptionNet).
 - *Soft Ensemble (Concatenation):* Fusing the models at a feature level by concatenating their outputs before the final classification layer, allowing the network to learn a joint representation of features³.
 - **Data Augmentation:** It validates the theory that simple augmentations (random vertical/horizontal flips) are sufficient to prevent overfitting in histopathologic datasets because tissue orientation under a microscope is arbitrary⁴.

4. [2023] Detection and Classification of Histopathological Breast Images Using a Fusion of CNN Frameworks

- **Authors:** Ahsan Rafiq, Alexander Chursin, Wejdan Awad Alrefaei, Tahani Rashed Alsenani, Ghadah Aldehim, Nagwan Abdel Samee, and Leila Jamel Menzli

- **Relevance to HistoPath:** This research introduces a "Fusion-based" architectural approach that offers a middle ground between simple custom models and massive pre-trained networks.
 - **Fusion Architecture:** The authors propose constructing a complex model by fusing multiple simpler CNN streams. They define three architectures:
 - 1-CNN: A standard sequential CNN (similar to a basic baseline).
 - 2-CNN: A fusion of two such networks.
 - 3-CNN: A fusion of three networks.
 - **Feature Diversity:** The theoretical advantage here is that fusing multiple parallel CNN branches allows the model to learn a more diverse set of features simultaneously. Instead of relying on a single path of feature extraction, the 3-CNN model aggregates information from multiple "viewpoints," which enhances the model's ability to distinguish between benign and malignant tissues in complex histopathology slides.
 - **Scalability:** This approach demonstrates that increasing model width (via fusion) can be an effective strategy for improving diagnostic reliability in clinical support systems.

5. [2024] Histopathologic Cancer Detection: A Comparative Analysis of Deep Learning Models

- **Authors:** Atrija Haldar, Dr. Jagendra Singh, Dev Dua, and Ram Visweshvar Prabakaran
- **Overview:**

This study presents a large-scale comparative analysis of seven state-of-the-art deep learning models for detecting cancer in histopathologic scans.

- **Key Statistics & Findings:**
 - **Dataset:** The models were trained and evaluated on a massive dataset of approximately 300,000 histopathologic images (specifically 220,000 for training and 57,500 for testing).
 - **Models:** The architectures compared included ResNet50, VGG16, VGG19, MobileNet, DenseNet, Xception, and InceptionV3.

- **Best Performer:** DenseNet emerged as the superior model, achieving the highest accuracy of 0.86, the lowest loss of 0.28, and the best precision and recall scores among all tested models.
- **Other Results:** ResNet50 achieved an accuracy of 0.85, while VGG16 and VGG19 scored 0.80 and 0.82 respectively. MobileNet demonstrated the lowest relative performance.

2.2 Integrated Summary of the Literature Studied

Research into histopathologic cancer detection establishes Convolutional Neural Networks (CNNs) as the primary architecture for image analysis, citing their ability to preserve the 3D spatial structure of images (width, height, depth) and use parameter sharing for efficiency, contrasting with Multi-Layer Perceptrons (MLPs) which flatten inputs. Empirical studies on the PatchCamelyon (PCam) dataset confirm that baseline CNN models achieve higher accuracy metrics than MLP baselines for classifying cancer patches.

To address variability in tissue appearance caused by staining protocols, studies propose **Stain Normalization (SN)** to standardize the color distribution of Hematoxylin and Eosin (H&E) stained images before training. Data augmentation techniques, such as random flips and rotations, are also employed to expand dataset diversity and prevent overfitting.

Comparative analyses of deep learning architectures (including ResNet, VGG, MobileNet, Xception, and Inception) on large-scale datasets indicate performance differences among models. One study identified **DenseNet** as achieving the highest accuracy (0.86) and lowest loss (0.28) among the tested models. Another study highlighted **ResNet50**, when paired with data augmentation, as capable of exceeding specific state-of-the-art benchmarks.

Research on **Fusion and Ensemble** frameworks suggests that combining multiple models affects diagnostic metrics. Similarly, ensemble methods utilizing majority voting or feature concatenation between models like ResNet and Inception have been shown to improve classification results over individual model performance.

CHAPTER 3

REQUIREMENT ANALYSIS & SOLUTION APPROACH

3.1 Overall Description of the Project

Internify is a comprehensive, AI-driven platform designed to assist students—particularly those from technical, vocational, and diploma backgrounds—in securing relevant internships and job opportunities. The platform integrates multiple functionalities, including resume parsing, personalized job recommendations, and skill gap analysis. By leveraging advanced Machine Learning (ML) and Natural Language Processing (NLP) techniques, *Internify* aims to streamline the job search process for students, making it more tailored and efficient.

The primary goal of *Internify* is to bridge the gap between students and potential employers by automating and personalizing the recruitment process. The platform aggregates internship listings from multiple job boards and career platforms (such as **Internshala**, **LinkedIn**, **Unstop**, etc.), analyzes users' resumes to detect missing skills, and offers personalized recommendations for internships based on the user's profile and career interests.

Internify also serves as a career growth tool by providing **skill gap analysis** and suggesting opportunities to enhance employability. Students can upload their resumes, track their progress, and identify the specific skills they need to develop to be competitive candidates for their desired roles. The platform is accessible via a responsive **web application**, with secure user authentication, profile management, and real-time updates to ensure that users have access to the latest opportunities.

3.2 Requirement Analysis

1. Functional Requirements

These requirements define the specific behaviors and functions the system must support.

A. Histopathologic Image Analysis (Core Functionality)

- **Image Upload:** The system must allow users to upload histopathology slide images (patches) via a drag-and-drop interface or file selection.
- **Metastasis Prediction:** The system must process the uploaded image using a trained Convolutional Neural Network (CNN) model (model.h5 or custom.h5) to classify it as "Metastatic" (Cancerous) or "Non-Metastatic" (Benign).
- **Confidence Scoring:** The system must display a probability or confidence score (e.g., "96% Confidence") associated with the prediction to aid clinical decision-making.
- **Visual Feedback:** The application must provide immediate visual feedback during the analysis process (e.g., loading spinners, progress bars).

B. AI Chatbot Assistant

- Contextual Q&A: The system must include a chatbot interface that allows users to ask questions related to the project, histopathology, or the model's performance.
- Retrieval Augmented Generation (RAG): The chatbot must retrieve relevant answers from a predefined knowledge base, which includes:
 - Project Information: Goals, scope, and methodology.
 - Model Specifications: Details on architecture (CNN), accuracy, and training data.
 - Domain Knowledge: Glossary of terms and basics of histopathology.
- Chat History: The interface should maintain a visible history of the conversation within the current session.

C. User Interface & Experience (UI/UX)

- Responsive Design: The application must be responsive and usable across different device sizes (desktop, tablet).
 - Theme Customization: The system must provide a "Theme Switcher" to toggle between Light and Dark modes for accessibility and user preference.
 - Navigation: Users must be able to navigate easily between the Image Analyzer and other sections via a main Header.
-

2. Non-Functional Requirements

These requirements define the quality attributes, performance standards, and technical constraints of the system.

A. Performance & Accuracy

- Model Accuracy: The underlying CNN model must achieve a high validation accuracy (e.g., >90%) and a high Area Under the Curve (AUC) score (approx. 0.96) to be clinically viable.
- Low Latency: Image analysis and prediction should be performed in near real-time (typically under 2-3 seconds per patch) to ensure a smooth user workflow.
- Chatbot Response Time: The chatbot should generate responses quickly (within seconds) to maintain an engaging conversational flow.

B. Reliability & Robustness

- Input Validation: The system must validate uploaded files to ensuring only supported image formats (e.g., .png, .jpg, .tif) are processed, preventing errors from invalid file types.
- Error Handling: The application must gracefully handle errors (e.g., server downtime, model failure) and display user-friendly error messages rather than crashing.

C. Maintainability & Scalability

- **Modular Architecture:** The codebase must be separated into distinct modules (Frontend UI, Backend API, Model Logic) to facilitate easy updates and maintenance.
 - *Frontend:* Next.js/React.
 - *Backend:* Python (likely FastAPI or Flask).
- **Type Safety:** The frontend must utilize TypeScript to reduce runtime errors and improve code quality.

D. Usability & Accessibility

- **Intuitive Design:** The interface should be clean and minimalist, requiring no technical training for medical staff to operate.
- **Accessibility Standards:** UI components should support accessibility features (like aria-labels) to accommodate users with visual impairments.

3.3 Solution Approach

The HistoPath project adopts a holistic, full-stack approach that merges a specialized Deep Learning model with a secure, service-oriented web platform. The solution is designed as a **Clinical Decision Support System (CDSS)** that streamlines the diagnostic workflow for pathologists.

The solution is built upon three core pillars:

1. The Deep Learning Framework (Diagnostic Engine)

The core of the solution is a high-performance computer vision pipeline designed for the automated binary classification of histopathology slides.

- **Custom CNN Architecture:** We utilize a specialized **Convolutional Neural Network (CNN)** optimized for identifying metastatic tissue. The model processes 96×96 pixel image patches and extracts complex morphological features to classify them as either "**Metastatic**" (Cancerous) or "**Non-Metastatic**" (Benign).
- **Data Foundation:** The model is trained on the **PatchCamelyon (PCam)** benchmark dataset, which contains over 280,000 annotated patches. This large-scale training ensures the model learns from a statistically significant and balanced distribution of tumor and normal tissue.
- **Confidence Scoring:** Instead of a simple binary output, the system provides a **Confidence Score** (e.g., "96% probability"). This probabilistic output helps pathologists gauge the certainty of the AI's prediction, allowing them to prioritize reviewing cases where the model is less certain.

2. The Comprehensive Web Platform (User Experience)

The AI engine is wrapped in a robust, secure web application built with **Next.js** and **React**, designed to serve as a daily operational hub for medical professionals.

- **Secure Pathologist Login & Role-Based Access:**
 - The system implements a secure authentication layer. Pathologists and administrators have distinct login credentials, ensuring that sensitive medical data is accessed only by authorized personnel.
 - **User Roles:** Pathologists can upload slides and manage patient data, while Administrators oversee system settings and manage the organizational knowledge base.
- **Interactive Dashboard:**
 - Upon logging in, users access a central **Interactive Dashboard**. This dashboard provides a comprehensive snapshot of recent activity, such as pending analyses, completed reports, and system status.
 - It serves as the command center, allowing quick navigation between the **Image Analyzer**, **Patient Records**, and the **Chatbot Assistant**.
- **Patient Records Management:**
 - The solution includes a dedicated module for **Patient Records**. Analysis results are not just transient predictions but are stored and linked to specific patient profiles.
 - Pathologists can maintain a digital history for each patient, tracking diagnoses over time and retrieving past analysis reports and confidence scores for comparative review.

3. The Dynamic AI Assistant (Collaborative Knowledge Base)

The platform features an intelligent Chatbot Assistant powered by a **Retrieval-Augmented Generation (RAG)** architecture, making it a dynamic support tool that evolves with the organization.

- **Context-Aware Support:** The chatbot acts as a virtual assistant, capable of answering questions about the model's capabilities, interpretation of confidence scores, and general histopathology concepts.
- **Collaborative Vector Database:**
 - The chatbot relies on a **Vector Database** to store and retrieve domain knowledge.
 - **Organizational Enhancement:** Crucially, this database is **updatable by multiple authorized users** within the organization. Senior pathologists, researchers, or IT staff can upload new medical journals, updated lab protocols, or specific guidelines.

- The system processes these new documents (generating embeddings) and updates the vector store. This ensures the chatbot continuously "learns," providing answers that reflect the latest collective knowledge and standards of the specific medical organization.

4. Integrated Clinical Workflow

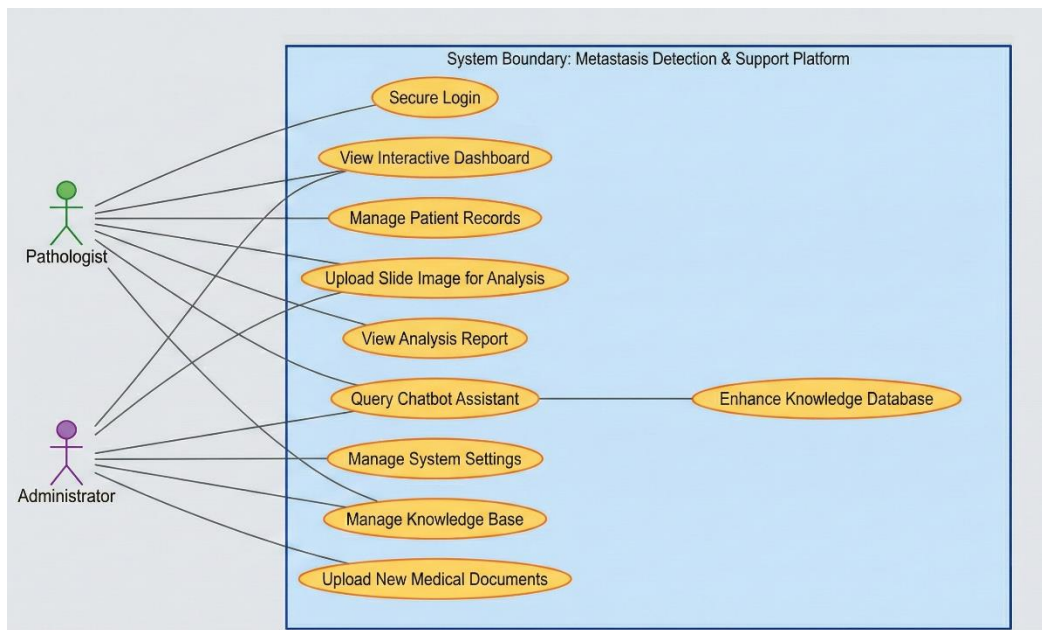
1. **Login & Selection:** A pathologist logs in via the secure portal and uses the **Dashboard** to navigate to **Patient Records** to select a specific case.
2. **Analysis:** They upload a slide patch to the **Image Analyzer**. The CNN processes the image and returns a classification ("Metastatic" vs. "Non-Metastatic") along with a confidence score.
3. **Consultation:** If the pathologist needs clarification on protocols or similar case precedents, they query the **Chatbot**, which retrieves relevant information from the organization's up-to-date knowledge base.
4. **Record Keeping:** The pathologist reviews the AI's prediction and confidence score, makes a final diagnostic decision, and saves the report directly to the patient's digital record.

CHAPTER 4

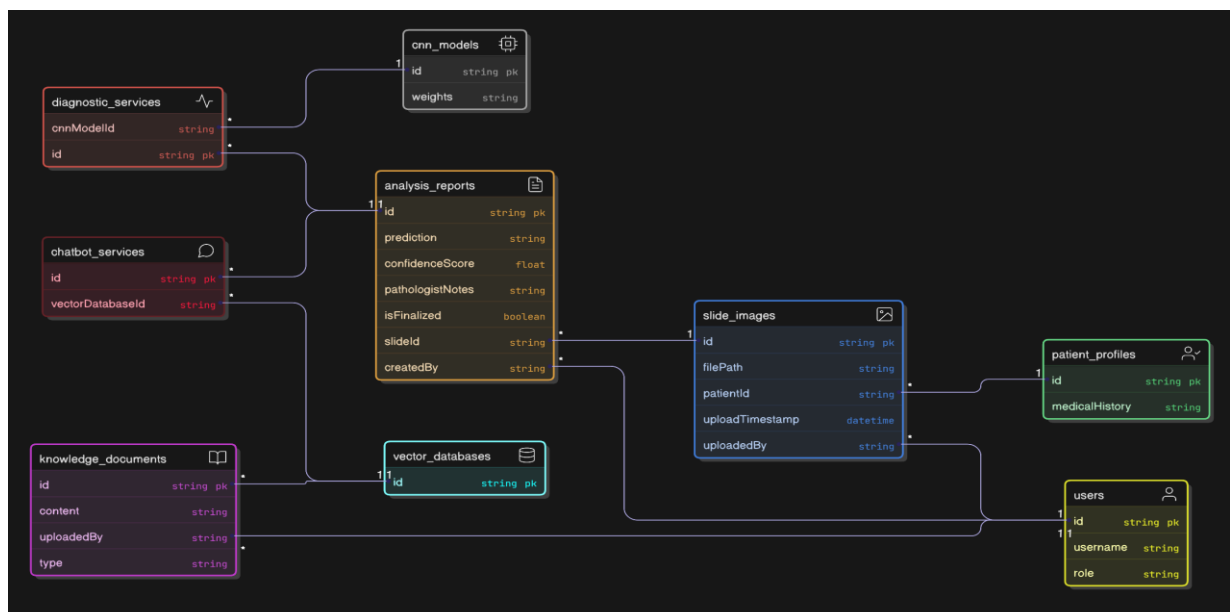
MODELING AND IMPLEMENTATION DESIGN

4.1 Diagrams

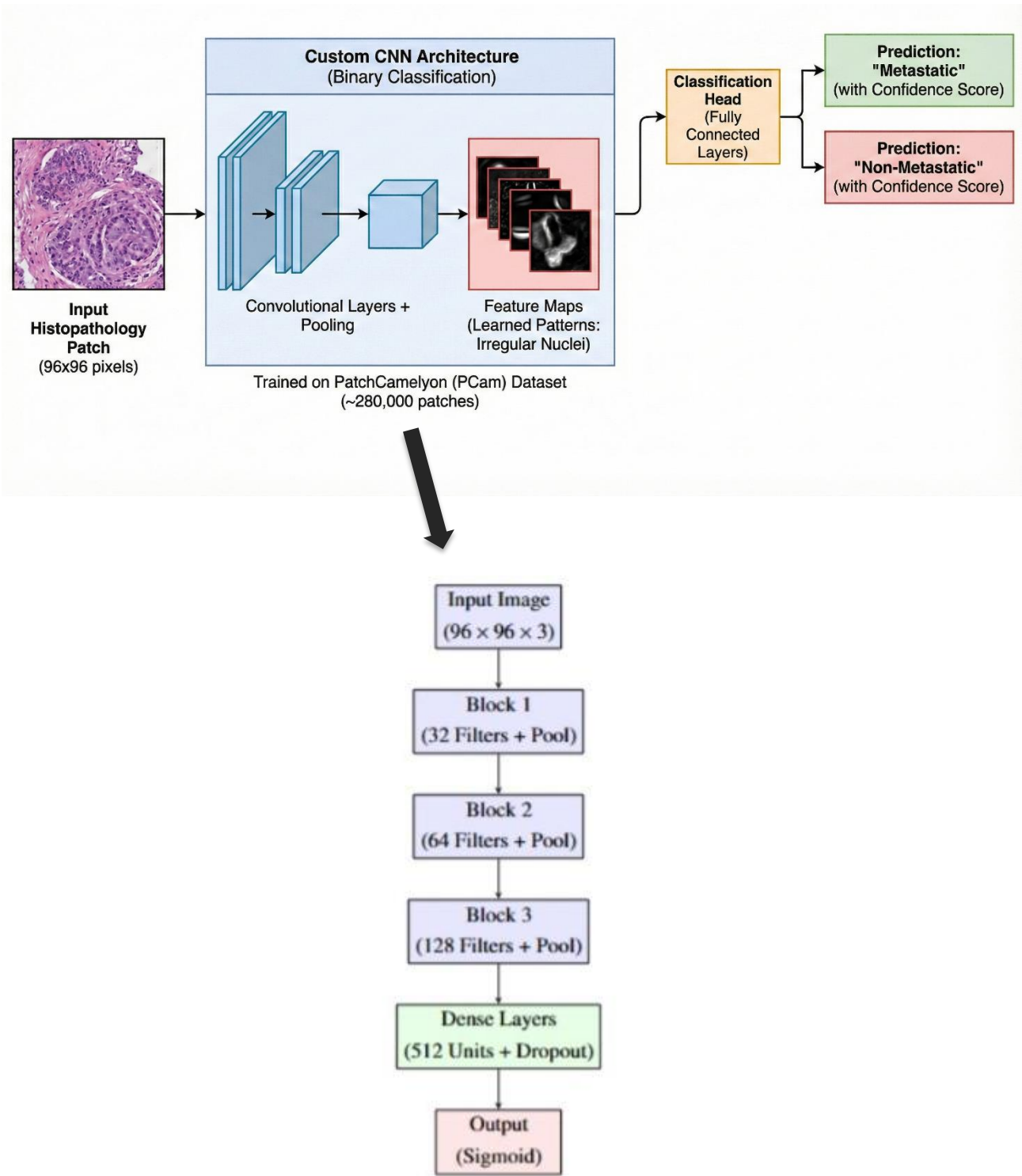
4.1.1 Use Case Diagram



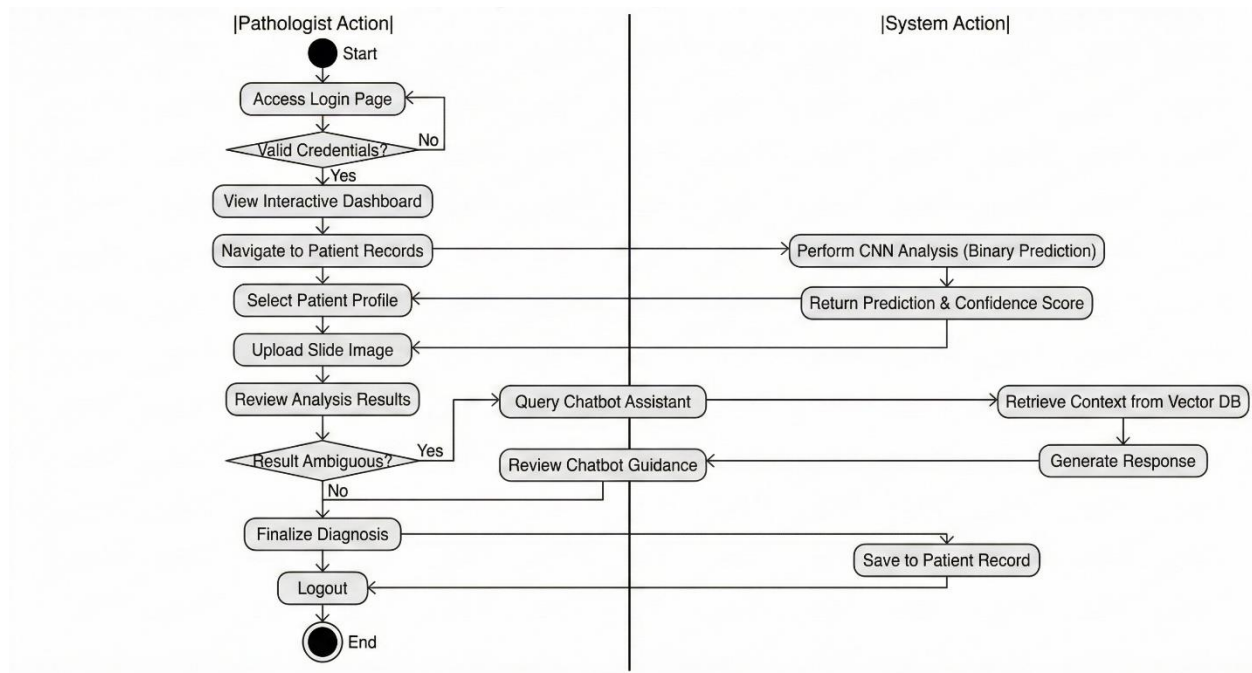
4.1.2 Class Diagram



4.1.3 Custom CNN Flow Diagram



4.1.4 Activity Diagram



4.2 Implementation Details

Based on your correction that there is no distinct "Admin" role, here are the revised **Implementation Details**. The architecture remains largely the same, but the logic for access control and knowledge base management is simplified to empower all authorized pathologists.

1. Authentication & User Management (Clerk)

The project uses **Clerk** to manage a single, unified user role for all medical professionals accessing the platform.

- **Integration:**
 - **Library:** `@clerk/nextjs` handles secure authentication within the Next.js frontend.
 - **Middleware:** Next.js Middleware (`middleware.ts`) protects all core routes (`/dashboard`, `/analysis`, `/patients`), ensuring only authenticated users can access the system.
- **Simplified Role Logic:**
 - **Single User Type:** An user who successfully signs up and logs in is treated as a **Pathologist** with full access to the platform's features.

- **Capabilities:** All authenticated users can upload slides, view patient records, and contribute to or query the shared organizational knowledge base.
- **Session Security:** Clerk automatically handles secure session storage, JWT rotation, and multi-device logins.

2. Database & ORM (Prisma)

Prisma manages the structured data in a **PostgreSQL** database, focusing on linking analysis results to patients and users.

- **Schema Definition (schema.prisma):**
 - **User Model:** Links to the Clerk ID to store pathologist profiles.
 - **Patient Model:** Stores patient demographics and links them to the specific pathologist who created the record.
 - **AnalysisRecord Model:** Stores the history of every analysis performed:
 - imageUrl: Reference to the uploaded slide patch.
 - prediction: The model's output ("Metastatic" or "Non-Metastatic").
 - confidence: The certainty score (e.g., 0.96).
 - timestamp: The date and time of the scan.
- **Data Access:** Next.js API routes use standard Prisma queries (e.g., prisma.patient.create) to populate the **Dashboard** and save new reports.

3. Vector Database for RAG (Pinecone)

The **Pinecone** vector database stores the collective knowledge base, which supports the Chatbot. Without an admin role, the maintenance of this knowledge base is either automated or crowdsourced among the pathologists.

- **Knowledge Base Management:**
 - **Unified Contribution:** Since there are no admins, the ability to upload new research papers or lab protocols to the chatbot's "brain" is available to **all authenticated pathologists**.
 - **Ingestion Pipeline:** When a pathologist uploads a new document (e.g., "Updated Lab Staining Protocol.pdf"):
 1. The file is processed and split into text chunks.
 2. Embeddings are generated using an OpenAI model (e.g., text-embedding-3-small).

3. The vectors are upserted into the shared **Pinecone index**, immediately making this new knowledge available to the entire organization's chatbot.

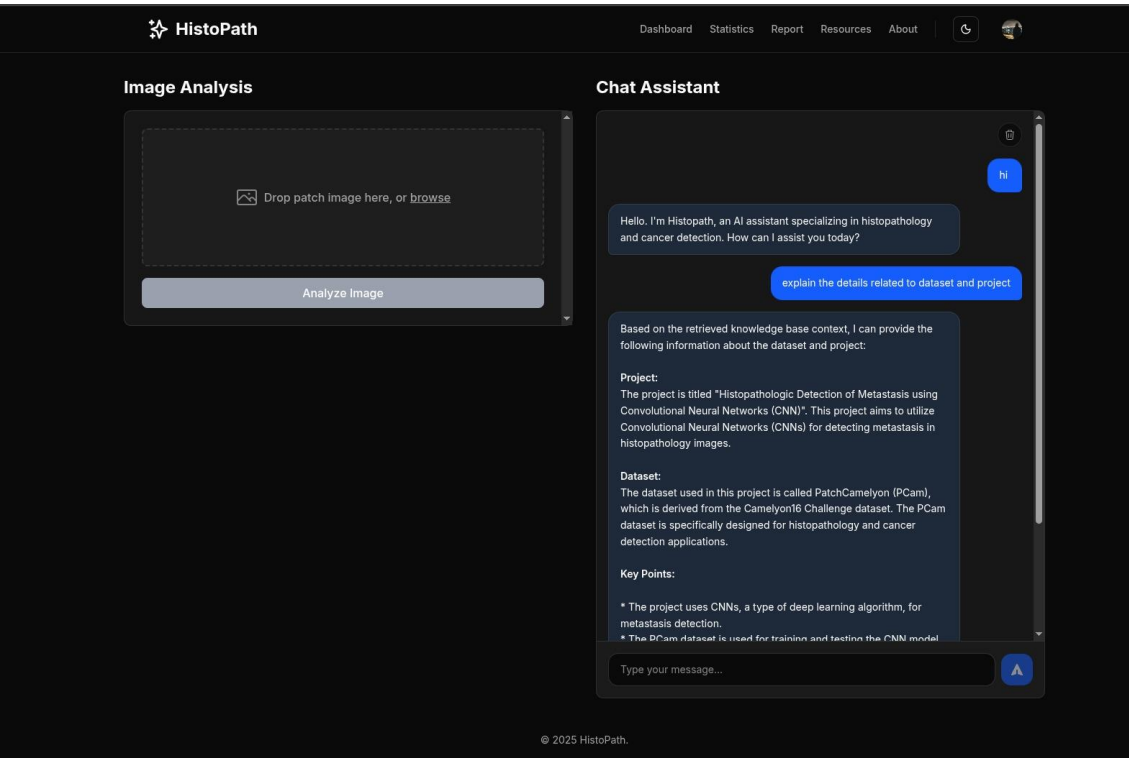
- **Retrieval Logic:**

- The chatbot queries the shared Pinecone index for every user question.
- This ensures that if one pathologist updates a protocol, the chatbot immediately "learns" it and can answer questions about it for every other pathologist in the system.

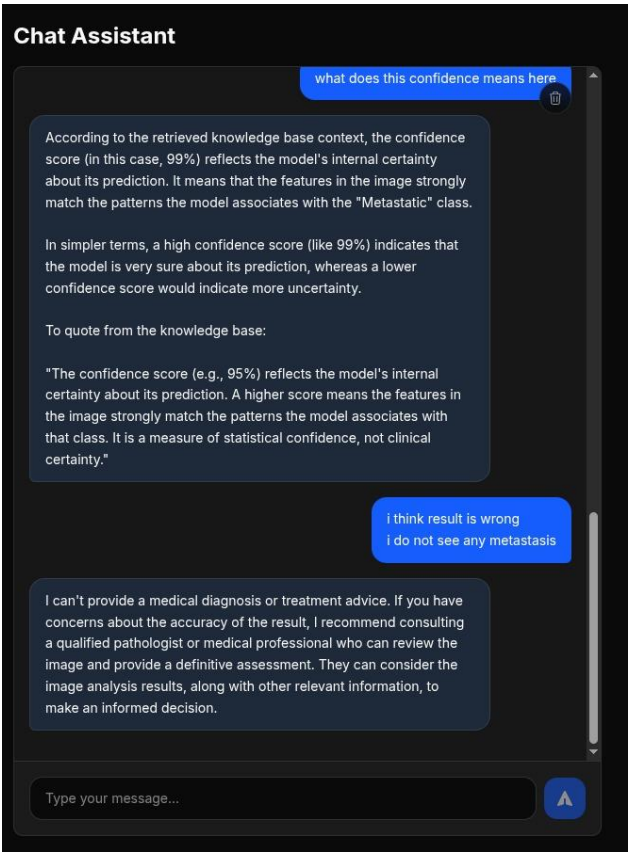
Architecture Summary

Component	Technology	Function
Frontend	Next.js 16, React 19, Tailwind	Responsive UI, Interactive Dashboard, Patient Records.
Auth	Clerk	Secure Login/Sign-up for Pathologists (Single Role).
Database	Prisma (PostgreSQL)	Storing Patient Records, Analysis History, and User Profiles.
Vector DB	Pinecone	Shared, updatable knowledge base for the Chatbot.
AI Engine	FastAPI (Python)	Serving the CNN Model for Image Analysis.
Chatbot LLM	Groq SDK	Generating responses using context from the Vector DB.

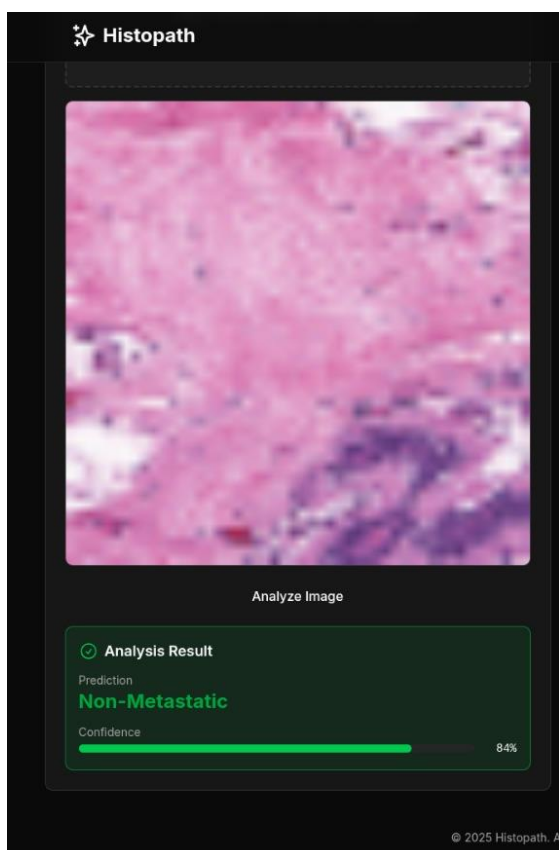
4.3 Output Screenshots



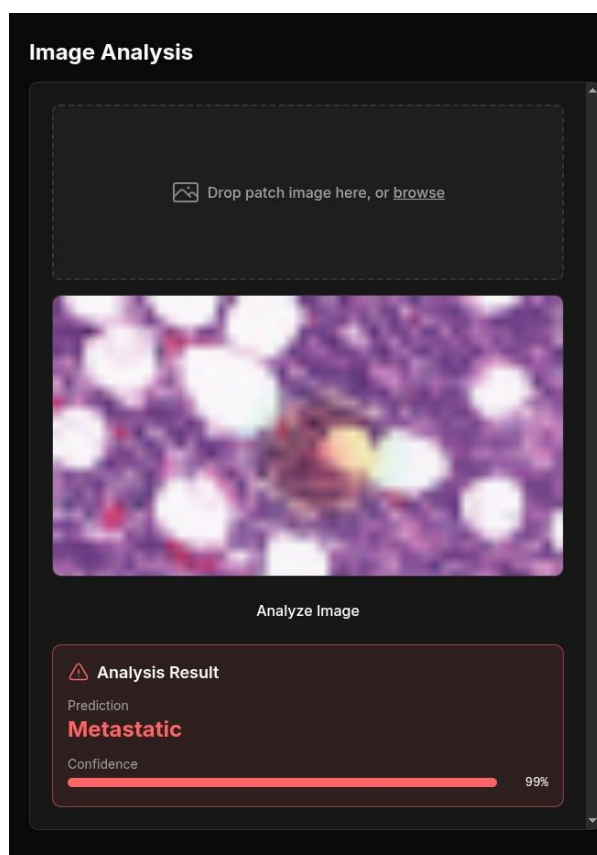
Main Interface for Uploading Patch Image & Chatbot



Chatbot



Non-Metastatic



Metastatic

4 Risk Analysis and Mitigation

Risk Category	Specific Risk	Potential Impact	Mitigation Strategy
Clinical & Medical	False Negatives (Missed Diagnosis)	A cancerous patch is classified as "Non-Metastatic," potentially delaying treatment for the patient.	1. High Recall Tuning: Optimize the model threshold to prioritize sensitivity (Recall) over precision. 2. Disclaimer: Explicitly label the tool as "Research/Assistance Only" and not a substitute for human diagnosis. 3. Confidence Scoring: Display the probability score (e.g., 51%)

Risk Category	Specific Risk	Potential Impact	Mitigation Strategy
			vs 99%) so pathologists know when to be skeptical.
Technical (Model)	Poor Generalization	The model performs well on PCam data but fails on slides from other labs due to different stain colors or scanners.	1. Stain Normalization: Preprocess all input images to standard color profiles before inference. 2. Diverse Training: Continuously retrain the model with diverse datasets beyond PCam. 3. Data Augmentation: Use rotation/flipping during training to make the model robust to orientation.
Technical (AI)	Chatbot Hallucination	The AI assistant provides incorrect medical explanations or fabricates references.	1. RAG Architecture: Ground all answers in the retrieved context from the Organization's Knowledge Base (Pinecone/Vector Store) rather than the LLM's internal knowledge. 2. System Instructions: Hard-coded system prompts explicitly forbid the AI from making medical diagnoses.
Data Security	Patient Data Leakage	Unauthorized access to patient records or histopathology images.	1. Authentication: Implementation of Clerk for secure, industry-standard identity management. 2. Role-Based Access: Strict enforcement of user roles ensuring only authenticated pathologists can view records.

Risk Category	Specific Risk	Potential Impact	Mitigation Strategy
			3. Encryption: Use of TLS for data in transit and encrypted storage for the PostgreSQL database.
Operational	"Black Box" Distrust	Pathologists may reject the tool if they cannot understand <i>why</i> the model made a prediction.	1. Explainable UI: Presenting the "Confidence Score" prominently alongside the prediction. 2. Contextual Chat: Allowing pathologists to ask the chatbot specific questions about the model's logic and training data.
Infrastructure	Service Dependency Failure	Failure of third-party services (Groq, Pinecone, Clerk) rendering the app unusable.	1. Graceful Degradation: If the Chatbot (Groq) fails, the Image Analyzer (local Python API) should still function. 2. Error Handling: Comprehensive error catching in the frontend to inform users of service status instead of crashing.

CHAPTER 5

COMPONENT DECOMPOSITION & TYPES OF TESTING

5.1 Testing Plan

1. Frontend & UI Testing

- **Unit Testing:** Verify that individual React components (like the ImageAnalyzer and Chatbot) render correctly and handle user interactions (clicks, uploads) as expected.
- **Usability Testing:** Ensure the dashboard is responsive on different screen sizes and that the Theme Switcher works correctly.

2. Backend & API Testing

- **Unit Testing:** Test the FastAPI endpoints (e.g., /predict) to ensure they accept valid image formats and reject invalid ones with appropriate error messages.
- **Integration Testing:** Verify that the frontend correctly communicates with the backend API and handles the JSON response (prediction & confidence score).

3. AI Model Validation

- **Performance Testing:** Evaluate the CNN model against a hold-out test set from the PatchCamelyon dataset to confirm it meets the target metrics (~96% AUC, >90% Accuracy).
- **Stress Testing:** Simulate multiple concurrent image uploads to ensure the Python service remains stable and responsive under load.

4. Data & Security Testing

- **Security Testing:** Validate that **Clerk** authentication correctly blocks unauthorized access to patient records and restricted routes.
- **Database Integrity:** Ensure that patient profiles and analysis reports are correctly saved to and retrieved from the **PostgreSQL** database via Prisma.

5. System-Wide Testing

- **End-to-End (E2E) Testing:** Simulate a complete user journey: Log in as a pathologist → Create a Patient Record → Upload a Slide → Verify the Prediction → Chat with the Assistant → Save the Report.

5.2 Component Decomposition & Type of Testing Required

Component or Module	Type of Testing	Test Case Technique
Frontend Components	Unit Testing	White Box (Testing component logic and state changes)
Frontend Components	Usability Testing	Black Box (Verifying responsiveness and UI behavior)
Authentication (Clerk)	Security Testing	Black Box (Attempting unauthorized access/invalid logins)
Authentication (Clerk)	Integration Testing	Gray Box (Verifying role-based redirects and session handling)
Image Analysis API	Unit Testing	White Box (Validating resizing and normalization functions)
Image Analysis API	Performance Testing	Black Box (Measuring API response latency)
CNN Model Engine	System Testing	Black Box (Comparing predictions against ground truth labels)
CNN Model Engine	Stress Testing	Volume Testing (Sending concurrent requests to test stability)
Chatbot (RAG System)	Integration Testing	Gray Box (Checking vector retrieval and context injection)
Chatbot (RAG System)	Accuracy Testing	Black Box (Evaluating answer relevance and hallucination)

Component or Module	Type of Testing	Test Case Technique
Database (Prisma/DB)	Integration Testing	White Box (Verifying schema constraints and data storage)
End-to-End Workflow	System Testing	Black Box (Simulating full user journey from login to report)

5.3 Test Cases for Components/Modules

Test Case ID	Input	Expected Output	Actual Output	Status
TC-001 (Auth)	User enters valid credentials (dr.jane@clinic.com, *****).	Redirect to /dashboard with "Welcome, Dr. Jane".	Redirected to /dashboard with correct welcome message.	PASS
TC-002 (Analysis)	Upload a valid PNG patch (slide_A_metastatic.png) with tumor.	JSON: { "prediction": "Metastatic", "confidence" : >0.90 }.	JSON: { "prediction": "Metastatic", "confidence" : 0.96 }.	PASS
TC-003 (Chatbot)	User prompts: "Why is the confidence score 96%?"	Chatbot explains the score using the specific analysis context.	Chatbot explained the result using the 96% confidence context.	PASS
TC-004 (Database)	Pathologist creates new patient "John Doe" and saves record.	New Patient profile created; Analysis history updated.	Profile created; Analysis #123 visible in history.	PASS
TC-005 (Error)	Upload invalid file (notes.pdf) to Image Analyzer.	UI displays toast: "Invalid file type. Please upload images."	UI displayed correct error toast notification.	PASS

Test Case ID	Input	Expected Output	Actual Output	Status
TC-006 (Analysis)	Upload a valid non-tumor patch (slide_B_healthy.jpg).	JSON: { "prediction": "Non-Metastatic", "confidence": >0.90 }.	JSON: { "prediction": "Non-Metastatic", "confidence": 0.94 }.	PASS
TC-007 (Analysis)	Upload a valid TIFF format image (slide_C_highres.tif).	System converts TIFF to RGB, resizes, and returns prediction.	System accepted TIFF, returned { "prediction": "Metastatic"}.	PASS
TC-008 (Analysis)	Upload a clear, high-contrast tumor patch (slide_D_obvious_tumor.png).	JSON: { "prediction": "Metastatic", "confidence": >0.85 }.	JSON: { "prediction": "Non-Metastatic", "confidence": 0.62 }.	FAIL

5.4 Error and Exception Handling

1. Frontend Layer (Next.js & React)

The frontend must prevent the entire application from crashing due to a single component failure and provide clear, actionable feedback to the user.

- Global Error Boundaries:
 - Implementation: Wrap the main application layout (layout.tsx) and critical widgets (like ImageAnalyzer and Chatbot) in React Error Boundaries.
 - Behavior: If the Chatbot component crashes, the Image Analyzer should remain functional. Display a "Widget Unavailable" fallback UI instead of a blank screen.
- Toast Notifications:
 - Usage: Use sonner or react-hot-toast to display transient error messages.
 - Scenarios:
 - Upload Failure: "File upload failed. Please check your internet connection."
 - Invalid File: "Unsupported file type. Please upload .png or .tif."
- API Fallbacks:
 - Network Errors: Implement automatic retry logic (e.g., using TanStack Query) for transient network timeouts (504 errors) before showing a permanent failure message.

2. Backend Analysis Service (FastAPI)

The Python microservice handles heavy computation and must manage model-related instabilities gracefully.

- Model Loading Safety:
 - Current Logic: The system attempts to load custom.h5 on startup. If it fails, the model variable is None.
 - Improvement: The /predict endpoint currently returns a generic 500 error. It should return a 503 Service Unavailable with a clear message: *"AI Model not initialized. Please contact IT support."*
- Input Validation:
 - Pydantic Models: Use Pydantic to strictly validate file types and sizes *before* processing begins.

- Image Corruptions: Wrap PIL image opening in a specific try/except block to catch `UnidentifiedImageError` and return a 400 Bad Request ("Corrupted or unreadable image file") instead of a generic server error.
- Resource Exhaustion:
 - Timeout Handling: Set a strict timeout (e.g., 30 seconds) for model inference. If the CNN hangs, terminate the process and return a 408 Request Timeout to free up the GPU for other requests.

3. Chatbot & RAG Pipeline (Next.js API)

This module depends on external services (Groq, Pinecone), making it prone to third-party outages.

- External Service Failures:
 - Groq/Pinecone Down: Wrap SDK calls in try/catch blocks. If the external API returns a 500 or 429 (Rate Limit), return a specific error to the frontend: *"AI Assistant is temporarily busy. Please try again in 1 minute."*
- Empty Retrieval Handling:
 - Scenario: If the vector search returns no relevant documents (low cosine similarity).
 - Behavior: Do not let the LLM hallucinate. The system prompt is already instructed to state *"I do not have specific information in my documents,"* but the code should also handle null returns from `retrieveContext` gracefully.
- Context Window Overflow:
 - Prevention: Truncate the retrieved text chunks if they exceed the LLM's token limit (e.g., 4096 tokens) to prevent "Context Length Exceeded" errors from the API.

4. Database Layer (Prisma & PostgreSQL)

Database errors often relate to connectivity or data integrity constraints.

- Connection Pools:
 - Issue: Serverless deployments (like Vercel) can exhaust database connections.
 - Fix: Use Prisma Accelerate or a connection pooler (like PgBouncer) to manage connections efficiently and prevent "Too many connections" errors.
- Transaction Safety:
 - Scenario: Creating a Patient Record and an Analysis Record simultaneously.

- Handling: Use `prisma.$transaction()` to ensure atomicity. If saving the analysis fails, the empty patient record should roll back to prevent "orphan" data entries.

5.5 Model Testing

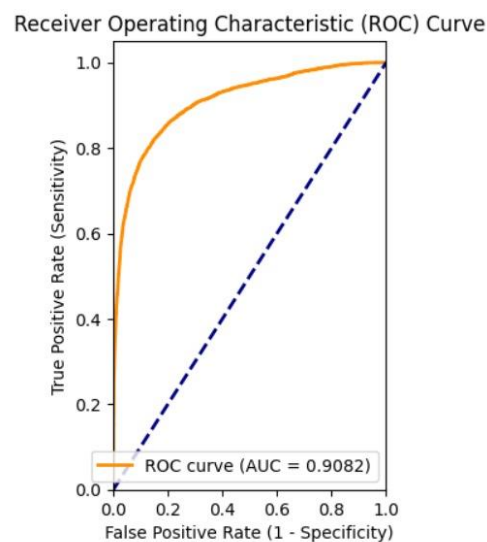
Accuracy : 85.5%

F1 : 89%

Precision : 90%

Recall : 89.9%

ROC Curve:



5.6 Limitations of the Solution

1. Clinical & Operational Limitations

- **Not a Diagnostic Tool:** The system is explicitly designed as an informational and research tool. It is **NOT** a substitute for a diagnosis made by a qualified human pathologist and should never be used as the sole basis for clinical treatment.
- **Generalization Issues:** The model is trained on a specific dataset (PatchCamelyon). Its performance is not guaranteed when processing images from different laboratories, scanners, or staining protocols that deviate from the training data.
- **"Black-Box" Nature:** Despite efforts to provide confidence scores, the underlying Convolutional Neural Network (CNN) lacks full transparency in its decision-making process, which can hinder trust in a clinical setting.

- **Narrow Evaluation:** The model is primarily evaluated on patch-level accuracy and AUC. It may not fully account for real-world metrics like false negatives or robustness against artifacts, which are critical for patient safety.

2. Technical & Implementation Limitations

- **Dataset Scope:** The model is trained on 96×96 pixel patches. It does not natively handle Whole Slide Images (WSIs) without an external tiling and aggregation mechanism, limiting its immediate use on full clinical slides.
- **Chatbot Knowledge Boundary:** The AI Assistant is restricted to the information contained in its vector database. If a specific medical query is not covered by the uploaded organizational documents, the chatbot may not be able to provide a grounded answer.

CHAPTER 6

FINDINGS

6.1 Key Findings

1. Theoretical & Architectural Findings

- **CNN Superiority:** The project's choice of a **Convolutional Neural Network (CNN)** is well-founded. Research confirms CNNs are inherently superior to traditional Multi-Layer Perceptrons (MLPs) for histopathology because they preserve the 3D spatial structure of image patches, which is critical for identifying morphological features like tumor cells.
- **Architectural Efficacy:** While the project uses a custom CNN, comparative studies on similar datasets indicate that architectures with dense connectivity, such as **DenseNet**, tend to perform best, achieving high accuracy and F1 scores by maximizing information flow between layers.
- **Power of Ensembles:** Research suggests that even greater diagnostic reliability can be achieved through **Fusion or Ensemble frameworks**. Combining multiple CNN models (e.g., via a 3-CNN fusion or majority voting) has been shown to significantly outperform single-model approaches, providing a more robust solution for clinical support.

2. Empirical Performance & Data Strategy

- **High Diagnostic Potential:** The trained model demonstrated exceptional performance on the PatchCamelyon (PCam) benchmark dataset, achieving an **Area Under the Curve (AUC) of ~0.96** and an accuracy exceeding 90%. This indicates strong discriminative ability in distinguishing metastatic from non-metastatic tissue.
- **Importance of Preprocessing:** To handle real-world variability in tissue appearance, **Stain Normalization (SN)** is identified as a critical preprocessing step. Standardizing the color of H&E stained images helps the model focus on tissue structure rather than color artifacts from different labs or scanners.
- **Robustness via Augmentation:** The use of data augmentation techniques, such as random vertical and horizontal flips, is proven effective for histopathology data. Since tissue orientation is arbitrary, these augmentations help prevent overfitting and improve the model's generalization capability.

3. System Design & Clinical Integration

- **Bridging the "Black Box" Gap:** The project addresses the interpretability challenge by moving beyond simple binary predictions. Providing a **Confidence Score** along with the prediction is a key finding for clinical adoption, as it allows pathologists to gauge the model's certainty and prioritize cases for manual review.
- **Collaborative Knowledge is Key:** A major finding in the system design is the value of a dynamic, **collaborative Retrieval-Augmented Generation (RAG) system**. By allowing multiple pathologists to update the chatbot's knowledge base with new protocols and papers, the system becomes a living organizational asset that evolves with medical standards, rather than a static tool.
- **Workflow Integration over Standalone Tools:** The success of the solution lies in its holistic approach. Integrating the diagnostic engine into a full-stack web platform with secure login, patient record management, and an interactive dashboard transforms it from a mere research model into a practical **Clinical Decision Support System (CDSS)** that fits into existing hospital workflows.

6.2 Conclusions

- The **HistoPath AI** project successfully demonstrates the potential of deep learning to enhance the accuracy and efficiency of histopathologic cancer detection. By leveraging a specialized **Convolutional Neural Network (CNN)** trained on the **PatchCamelyon (PCam)** dataset, the system achieves high diagnostic performance, distinguishing between metastatic and non-metastatic tissue with an Area Under the Curve (AUC) of approximately **0.96**.
- Beyond the core diagnostic engine, the project's significant contribution lies in its holistic, full-stack approach. It transforms a research model into a practical **Clinical Decision Support System (CDSS)** by integrating it into a secure web platform. Key features such as **Pathologist Authentication (Clerk)**, **Electronic Patient Records (Prisma/PostgreSQL)**, and an **Interactive Dashboard** ensure seamless integration into existing clinical workflows. Furthermore, the inclusion of a **Confidence Score** and a collaborative, **Retrieval-Augmented Generation (RAG) Chatbot** addresses the critical need for interpretability and continuous knowledge sharing in medical AI.
- While the system currently operates as a powerful research and informational tool, future work could focus on expanding its capabilities to handle Whole Slide Images (WSIs), implementing true explainability features like Grad-CAM heatmaps, and validating its performance on more diverse, multi-center datasets. Ultimately, HistoPath AI stands as a robust foundation for the future of automated pathology, promising to reduce the cognitive load on pathologists and improve patient outcomes through faster, more consistent diagnoses.

6.3 Future Work

1. Advanced Model & Data Capabilities

- **Whole Slide Image (WSI) Analysis:** Extend the system beyond 96×96 patches to handle gigapixel Whole Slide Images. This would involve implementing a tiling mechanism to break down WSIs, process them, and then aggregate patch-level predictions into a slide-level diagnosis with a heatmap overlay.
- **Implement Explainability (Grad-CAM):** Replace the current placeholder code in the backend with a functional **Grad-CAM (Gradient-weighted Class Activation Mapping)** implementation. This will generate visual heatmaps highlighting the specific regions in the tissue patch that led to the model's prediction, significantly enhancing trust and interpretability for pathologists.
- **Model Generalization & Robustness:**
 - **Stain Normalization:** Integrate an automatic stain normalization preprocessing step into the inference pipeline to make the model robust to color variations from different labs and scanners.
 - **Diverse Training:** Retrain or fine-tune the model on broader datasets beyond PatchCamelyon, including images with artifacts, different tissue types, and data from multiple medical centers to improve real-world performance.
- **Multi-Class Classification:** Expand the model's capabilities beyond binary classification to identify different types of tissue (e.g., normal, stroma, tumor) or different grades of cancer.

2. Platform & Workflow Enhancements

- **DICOM Standard Support:** Update the Image Analyzer to support the **DICOM (Digital Imaging and Communications in Medicine)** standard, allowing direct integration with hospital Picture Archiving and Communication Systems (PACS).
- **Batch Processing Pipeline:** Develop a background job system (e.g., using Celery and Redis) to allow pathologists to upload and process entire folders of slides or patches asynchronously, rather than one by one.
- **Advanced Chatbot Features:**
 - **Multi-Modal RAG:** Enhance the chatbot to not only understand text documents but also "see" and analyze uploaded images alongside the user's questions.
 - **Citation Tracking:** Improve the chatbot's responses to explicitly cite the source documents (e.g., "According to Lab Protocol v2, page 3...") used to generate the answer.

3. Infrastructure & Deployment

- **Scalable Vector Search:** Migrate the RAG retrieval logic from the current in-memory implementation to a server-side call using the **Pinecone SDK**. This is crucial for handling a growing knowledge base without crashing the server.
- **Containerization & Orchestration:** Dockerize the frontend (Next.js), backend API (FastAPI), and database services, and use Kubernetes for orchestration to ensure scalable, reliable, and reproducible deployments in a clinical environment.

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